

SMARTBRIDGE / SKILL WALLET

Solution Architecture Blueprint

Date: 2 July 2026

Team ID: —

Project Name: Caption AI : AI Powered Social Media Caption Generator

Solution Architecture Diagram

Presentation / Client Layer: HTML5 Dashboard

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Auth Service: .env Token Secrets

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API Gateway: Ngrok Proxy Tunnel

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**Core Logic: Python Flask & Groq Cloud LLM
Pipeline**



Data / Storage Layer: Local .env Config Maps

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Manages user input fields, target profile selection, and displays generation results.	HTML5, CSS3, JavaScript
API Gateway	Forwards traffic from local network tunnels to the backend routing engine.	Ngrok Proxy Agent
Microservices	Validates requests, manages Flask routing logic, and interfaces with the LLM.	Python Flask, Groq API
Database	Handles secure configuration storage and transient runtime token keys.	Local .env File Management